Baoding Wang

PhD student | School of Power and Mechanical Engineering | Wuhan University

Laser ultrasonics · Non-destructive evaluation · Physics-informed wavefield analysis

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RESEARCH PROFILE

I develop non-contact laser-ultrasonic measurements for composite interfaces and additively manufactured materials, combining experimental systems, multiphysics modelling and physics-informed learning. My work spans air-coupled inspection, sparse wavefield reconstruction, measurement-bias correction and in-process laser-cleaning monitoring.

EDUCATION

Wuhan University | PhD student | In progress

Guangdong University of Technology | MEng, Mechanical Engineering | 2019.09 - 2022.06

Thesis: Air-coupled laser-ultrasonic testing of microdefects in layered composites.

Engineering and Technology College, Hubei University of Technology
BEng, Mechatronic Engineering | 2014.09 - 2018.06

RESEARCH AND ENGINEERING CAPABILITIES

Experimental systems: laser-ultrasonic platform design; laser/galvanometer/acquisition synchronization in LabVIEW; motion control, fixture design and optical alignment.

Modelling and analysis: COMSOL multiphysics simulation; ultrasonic wavefield processing; physics-informed neural networks; Python, PyTorch and MATLAB.

Software engineering: Java and Python development, requirements analysis, technical design, implementation, debugging and cross-team integration.

PROFESSIONAL EXPERIENCE

Midea Group - Software Engineer
Software requirements, design, implementation, debugging and deployment support.

Han's Laser - Mechanical Designer | 2018.09 - 2019.08
Laser-equipment design, component selection and prototype assembly.

HONORS, PATENTS AND COMMUNICATION

Outstanding Graduate, Guangdong University of Technology (2022); academic scholarships (2019-2021); related research presented at the 2021 Far East NDT forum.

Contributing inventor on two granted Chinese invention patents: 202010897292.9 (laser-ultrasonic testing of woven carbon-fiber composites) and 202010896006.7 (automated laser-ultrasonic inspection of composite blades).

Baoding Wang | Publications

7 published journal articles and 1 accepted article. Equal contribution is marked with a dagger in the website bibliography.

1. Physics-informed broadband measurement of compliant interfacial defects under severe solid–air impedance mismatch using air-coupled laser ultrasonics
Baoding Wang, Lvming Zeng, Xuanrong Ji, Jun Zhang.
Measurement 290, 122975 (2026).
First author | DOI: 10.1016/j.measurement.2026.122975

2. Process-integrated self-sensing pulsed laser cleaning with ultrasonic monitoring for Surface-Integrity indication and abnormality mapping
Baoding Wang, Tongxin Wang, Huakai Mao, Guodong Zhang, Jun Zhang.
Optics & Laser Technology 204, 116126 (2026).
Co-first author | DOI: 10.1016/j.optlastec.2026.116126

3. Sparse wavefield reconstruction and defect indication in low-SNR laser ultrasonics using physics-informed neural networks
Baoding Wang, Jiuzhang Li, Haodong Chen, Liequan Wu, Pan Ma, Bin Yang, Jun Zhang.
Ultrasonics 168, 108194 (2026).
Co-first author | DOI: 10.1016/j.ultras.2026.108194

4. Decoupling propagation and texture-induced artifacts to improve time-of-flight reliability in scanning laser-ultrasonic measurements of woven CFRP
Baoding Wang, Haodong Chen, Liequan Wu, Jiuzhang Li, Pan Ma, Bin Yang, Jun Zhang.
Measurement 279, 121669 (2026).
First author | DOI: 10.1016/j.measurement.2026.121669

5. Non-contact detection of the interfacial microdefects in metal/CFRP hybrid composites using air-coupled laser ultrasound
Baoding Wang, Zhongwen Cheng, Weisheng Liao, Bainian Long, Junwei Wu, Lvming Zeng, Xuanrong Ji.
Structural Health Monitoring 24(1), 243–254 (2025). First published online in 2024.
Co-first author | DOI: 10.1177/14759217241234557

6. Air-coupled laser ultrasonic testing of glass fiber reinforced polymer defect [in Chinese]
Wang B., Wu J., Liao W., Cheng Z., Deng Y., Zhang Y., Zeng L., Ji X..
Nondestructive Testing 44(6), 26–31 (2022).
First author | DOI: 10.11973/wsjc202206006

7. High-resolution air-coupled laser ultrasound imaging of microstructure and defects in braided CFRP
Lvming Zeng, Baoding Wang, Xu Liu, Junwei Wu, Lijun Deng, Maodan Yuan, Yan Chen, Yu Deng, Yongkang Zhang, Xuanrong Ji.
Composites Communications 28, 100915 (2021).
Second author | DOI: 10.1016/j.coco.2021.100915

8. Bias-Compensated Air-Coupled Laser-Ultrasonic Framework for Noncontact 2.5D Topography Measurement of Rough Additively Manufactured Surfaces
Baoding Wang, Liequan Wu, Tongxin Wang, Fang Dong, Bin Yang, Jun Zhang.
IEEE Transactions on Instrumentation and Measurement (2026). **Accepted; DOI available.**
First author | DOI: 10.1109/TIM.2026.3744057